might be suggested by the following report from the magazine *New Scientist* about changes in levels of the masculine hormone testosterone: 'Levels doubled in tennis players during the 24 hours before a big match. Afterwards, the levels in winners stayed up, but in losers they dropped.'

***p. 109* … the ESS concept as one of the most important advances in evolutionary theory since Darwin.**

This sentence is a bit over the top. I was probably over-reacting to the then prevalent neglect of the ESS idea in the contemporary biological literature, especially in America. The term does not occur anywhere in E. O. Wilson's massive *Sociobiology*, for instance. It is neglected no longer, and I can now take a more judicious and less evangelical view. You don't actually *have* to use ESS language, provided you think clearly enough. But it is a great aid to thinking clearly, especially in those cases—which in practice is most cases—where detailed genetic knowledge is not available. It is sometimes said that ESS models assume that reproduction is asexual, but this statement is misleading if taken to mean a positive assumption of asexual as opposed to sexual reproduction. The truth is rather that ESS models don't bother to commit themselves about the details of the genetic system. Instead they assume that, in some vague sense, like begets like. For many purposes this assumption is adequate. Indeed its vagueness can even be beneficial, since it concentrates the mind on essentials and away from details, such as genetic dominance, which are usually unknown in particular cases. ESS thinking is most useful in a negative role; it helps us to avoid theoretical errors that might otherwise tempt us.

***p. 113* Progressive evolution may be not so much a steady upward climb as a series of discrete steps from stable plateau to stable plateau.**

This paragraph is a fair summary of one way of expressing the now well-known theory of punctuated equilibrium. I am ashamed to say that, when I wrote my conjecture, I, like many biologists in England at the time, was totally ignorant of that theory, although it had been published three years earlier. I have since, for instance in *The Blind Watchmaker*, become somewhat petulant—perhaps too much so—over the way the theory of punctuated equilibrium has been oversold. If this has hurt anybody's feelings, I regret it. They may like to note that, at least in 1976, my heart was in the right place.